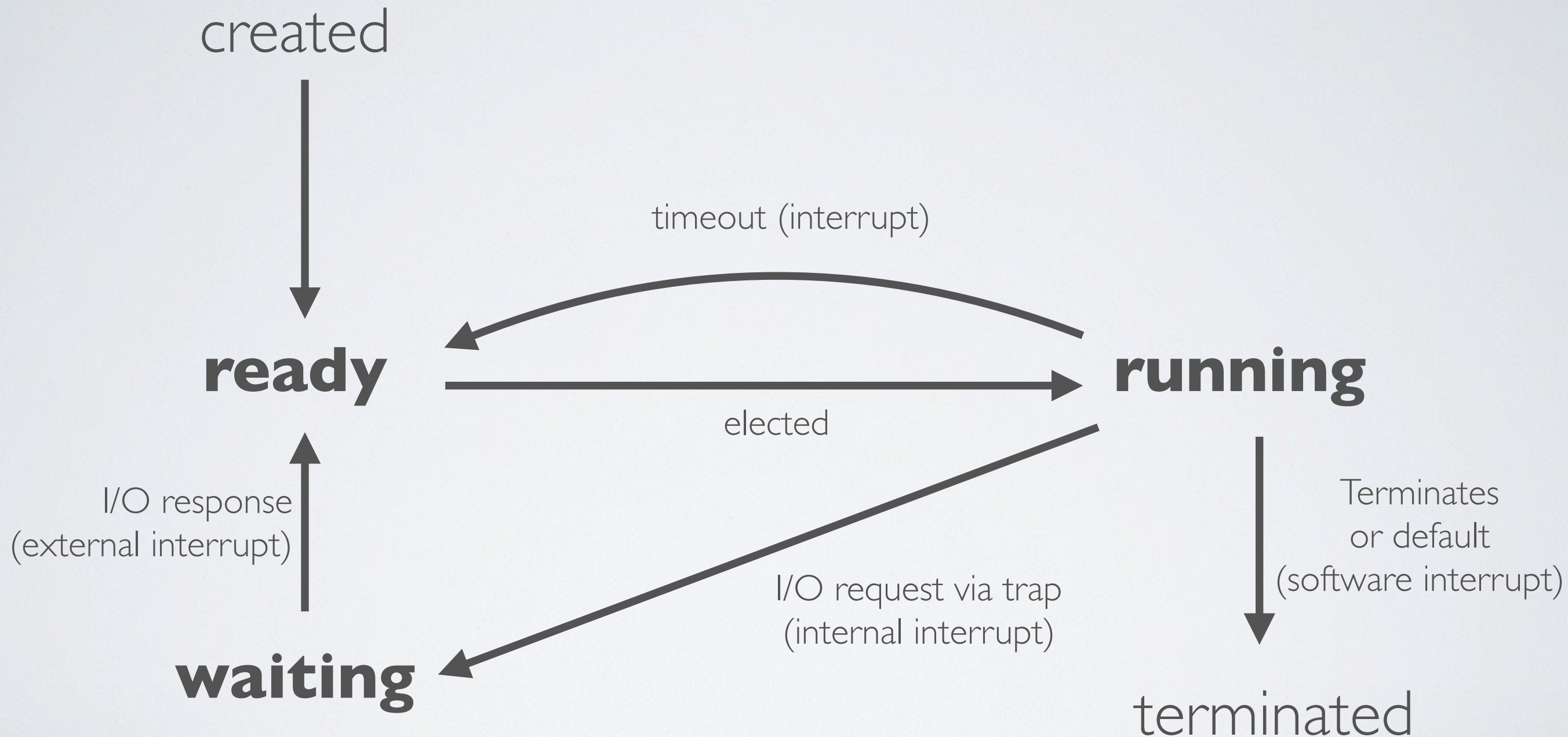


The different states of a thread



Context switching when

When the OS receives a fault

1. suspends the execution of the running thread
2. terminate the thread

When the OS receives a System Clock Interrupt or a System Call Trap (I/O request)

3. suspends the execution of the running thread
4. saves its execution context
5. changes the thread's state to ready (timeout) or waiting (I/O request)
6. elects a new thread from the ones in the ready state
7. changes its state to running
8. restores its execution context
9. resumes its execution

When the OS receives any other I/O interrupt

1. executes the I/O operation
2. switches the thread, that was waiting for that I/O operation, into the ready state
3. resumes the execution of the current program

→ **For each thread, the OS needs to keep track of its state (ready, running, waiting) and its execution context (registers, stack, heap and so on)**